

Properties of the Me Operation and Me-Scalar Multiplication on Elliptic Curves over Finite Fields

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Abstract

The M operation was introduced by Yura as an alternative to the max operation appearing in the box-ball system (BBS) to construct a BBS over finite fields. The Me operation is a version of the M operation for an elliptic curve E over a finite field $\mathbb{F}_p$. As with the M operation, the Me operation satisfies the idempotent law and does not satisfy the associative law. Nevertheless, for $P, Z \in E(\mathbb{F}_p)$ and $n \in \mathbb{N}$, the 1st Me-scalar multiplication $P_{n,Z}^I$ with auxiliary element Z can be defined. Moreover, for $P, Z \in E(\mathbb{F}_p)$ and $n \in \mathbb{Q}_+$, the 2nd Me-scalar multiplication $P_{n,Z}^{II}$ with auxiliary element Z can be defined. This paper shows the following properties that may be useful to construct cryptographic protocols: $(P_{n_0,Z}^I)_{n_1,Z}^I = (P_{n_1,Z}^I)_{n_0,Z}^I$, $(P_{n_0,Z}^{II})_{n_1,Z}^{II} = (P_{n_1,Z}^{II})_{n_0,Z}^{II} = P_{n_0 n_1, Z}^{II}$; the 1st MeDLP and the 2nd MeDLP, which are Me versions of the ECDLP, are difficult to solve on classical computers under certain conditions; the 1st MeCDH and the 2nd MeCDH, which are Me versions of the ECCDH, are NOT difficult to solve; and the sequence $\{P_{n,Z}^{II} : n = 1, 2, 3, \dots\}$ is nonperiodic unless it is constant.

Keywords: Finite Field, Elliptic Curve, Me Operation, Discrete Logarithm Problem, Nonperiodicity

1 Introduction

RSA and elliptic curve cryptosystems (ECCs) are currently widely used public-key cryptosystems. RSA cryptography is based on the hardness of integer factorization for its security, while ECC is based on the hardness of the elliptic curve discrete logarithm problem (ECDLP). However, Shor's algorithm [1] running on a quantum computer can solve integer factorization and the ECDLP in a short time, and the realization of large-scale quantum computers would compromise the security of RSA and ECC. Therefore, post-quantum cryptography, which is resistant to attacks using quantum computers, has been actively studied. In particular, the National Institute of Standards and Technology has been developing standards for post-quantum cryptography [2]. Representative types of post-quantum cryptography include lattice-based cryptography, isogeny-based cryptography, and multivariate cryptography.

Shor's algorithm is a quantum algorithm that uses the quantum Fourier transform to find the period of a periodic function over a group. Therefore, it is interesting to consider analogues of the discrete logarithm problem based on algebraic structures different from conventional groups.

The M operation [3] was introduced by Yura as an alternative to the max operation appearing in the box-ball system (BBS) to construct a BBS over finite fields. The Me operation is a version of the M operation for elliptic curves over finite fields. As with the M operation, the Me operation satisfies the idempotent law and does not satisfy the associative law. However, for $P, Z \in E(\mathbb{F}_p)$ and $n \in \mathbb{N}$, the 1st Me-scalar multiplication $P_{n,Z}^I$ with auxiliary element Z can be defined. Moreover, for $P, Z \in E(\mathbb{F}_p)$ and $n \in \mathbb{Q}_+$, the 2nd Me-scalar multiplication $P_{n,Z}^{II}$ with auxiliary element Z can be defined.

The author began studying the Me operation with the aim of applying it to cryptography [4]–[14]. In the course of this research, a cycle of discovering new properties of the Me operation, constructing cryptographic protocols based on conjectured computational hardness, and finding that the conjectured problems were not difficult was repeated.

To the best of the author’s knowledge, apart from numerous presentations at domestic conferences and workshops, only two works on the Me operation have been published in other venues: [7] and [15]. [7] is an early work by the author from the beginning of his research on the Me operation and concerns properties of the Me operation, including the proposal of a cryptographic protocol based on a conjecture that was later shown to be incorrect. [15] concerns pseudorandom number generation using the Me operation by Yamauchi et al.

The purpose of this paper is not to propose cryptographic protocols using the Me operation, but to show properties of the Me operation from the viewpoint of discrete mathematics, particularly the following properties that may be useful for constructing cryptographic protocols in the future: $(P_{n_0,Z}^I)_{n_1,Z}^I = (P_{n_1,Z}^I)_{n_0,Z}^I$, $(P_{n_0,Z}^{II})_{n_1,Z}^{II} = (P_{n_1,Z}^{II})_{n_0,Z}^{II} = P_{n_0 n_1,Z}^{II}$; the 1st MeDLP and the 2nd MeDLP, which are Me versions of the ECDLP, are difficult to solve on classical computers under certain conditions; the 1st MeCDH and the 2nd MeCDH, which are Me versions of the ECCDH, are NOT difficult to solve; and the sequence $\{P_{n,Z}^{II} : n = 1, 2, 3, \dots\}$ is nonperiodic unless it is constant.

The remainder of this paper is organized as follows. Section 2 introduces the M operation and elliptic curves. Section 3 proposes a list-free M operation. Section 4 discusses the Me operation and related concepts, particularly Me-scalar multiplication and the Me version of the ECDLP. Section 5 concludes this paper and presents directions for future research. The appendix defines M-scalar multiplication and the MDLP and discusses their properties.

1.1 Symbols

Throughout this paper, $\mathbb{N}$ and $\mathbb{Q}_+$ denote the sets of positive integers and positive rational numbers, respectively. For a prime p , $\mathbb{F}_p$ denotes the finite field of order p . For a finite set S , $\#S$ denotes the number of elements in S .

2 Preliminaries

In this section, we introduce the properties of the M operation over finite fields and elliptic curves that are needed in this paper.

2.1 M operation over finite fields

For a prime $p \geq 3$, Yura proposed the M operation $M : \mathbb{F}_p \times \mathbb{F}_p \rightarrow \mathbb{F}_p$, defined as follows [3].

- 1) Let $l = (p - 1)/2$, and choose l distinct elements $\alpha_1, \alpha_2, \dots, \alpha_l \in \mathbb{F}_p$.
- 2) Define $M(x, 0)$ as follows.

$$\begin{aligned} M(0, 0) &= 0, \\ M(x, 0) &= \sum_{j=1}^l (\delta_{x,j} \alpha_j + \delta_{x,-j} (\alpha_j - j)) \\ &= \begin{cases} \alpha_x & \text{if } 1 \leq x \leq l \\ \alpha_{p-x} - j & \text{if } l < x \leq p - 1, \end{cases} \end{aligned}$$

where, δ is the Kronecker delta.

- 3) For any $a, b \in \mathbb{F}_p$, define

$$M(a, b) = M(a - b, 0) + b.$$

The value of $M(a, b)$ depends on the choice of $\alpha_1, \alpha_2, \dots, \alpha_l$, and, in general, a list of $\alpha_1, \alpha_2, \dots, \alpha_l$ is required to compute $M(a, b)$. In this paper, we denote $M(a, b)$ by $a \oplus b$.

The M operation has the following properties.

Proposition 1.

- (a) $a \oplus a = a$. (*Idempotent law*)
- (b) $a \oplus b = b \oplus a$. (*Commutative law*)
- (c) $(a \oplus b) + c = (a + c) \oplus (b + c)$. (*Distributive law*)
- (d) In general, $(a \oplus b) \oplus c \neq a \oplus (b \oplus c)$. (*Nonassociativity*)

Proof. Properties (b), (c), and (d) are given in [3]. Property (a) follows immediately from the definition of the M operation. $\square$

2.2 Elliptic Curves

For a field $\mathbb{K}$, let E be an elliptic curve over $\mathbb{K}$ given in short Weierstrass form by

$$E/\mathbb{K} : y^2 = x^3 + ax + b, \quad a, b \in \mathbb{K}, \quad 4a^3 + 27b^2 \neq 0. \quad (1)$$

The set $E(\mathbb{K})$ of $\mathbb{K}$ -rational points of E is defined by

$$E(\mathbb{K}) = \{(x, y) \in \mathbb{K} \times \mathbb{K} : (x, y) \text{ satisfies (1)}\} \cup \{\mathcal{O}\},$$

where $\mathcal{O}$ is the point at infinity. For any $P, Q \in E(\mathbb{K})$, $P + Q \in E(\mathbb{K})$ is defined, and it is known that $(E(\mathbb{K}), +)$ forms an abelian group with $\mathcal{O}$ as the identity [16]. For a natural number n and $P \in E(\mathbb{K})$, the scalar multiplication

$$nP = P + P + \dots + P \quad (n \text{ terms})$$

is defined. The smallest $l \in \mathbb{N}$ satisfying $lP = \mathcal{O}$ is called the order of P . If $\mathbb{K}$ is a finite field, then $(E(\mathbb{K}), +)$ forms a finite abelian group.

For $P, Q \in E(\mathbb{F}_p)$ and the order l of P , the problem of finding $n \in \mathbb{Z}/l\mathbb{Z}$ satisfying $Q = nP$ is called the elliptic curve discrete logarithm problem (ECDLP). In this paper, we denote this problem by $\text{ECDLP}(P, Q)$. For $P \in E(\mathbb{F}_p)$, $Q_1 = nP$, and $Q_2 = mP$, the problem of computing nmP is called the elliptic curve computational Diffie-Hellman problem (ECCDH problem). In this paper, we denote this problem by $\text{ECCDH}(P, Q_1, Q_2)$.

Except in special cases, if $\#E(\mathbb{F}_p)$ has a sufficiently large prime factor (e.g., a 256-bit prime factor), the ECDLP and the ECCDH problem are believed to be difficult to solve with classical computers. However, if a sufficiently large quantum computer is realized, Shor's algorithm can solve the ECDLP and the ECCDH problem in polynomial time with respect to the bit length of $\#E(\mathbb{F}_p)$.

3 List-Free M Operation

For an odd prime p , let $l = (p - 1)/2$. When p is sufficiently large, it is difficult to keep the list $\{\alpha_1, \alpha_2, \dots, \alpha_l\}$ required to compute the M operation. Therefore, we would like to compute the M operation without using the list.

In this section, we assume that the prime p satisfies $p \equiv 3 \pmod{4}$. Then, for the Legendre symbol $(\cdot)$,

$$\left(\frac{-1}{p}\right) = -1. \quad (2)$$

Proposition 2 ([4]).

Let p be a prime satisfying $p \equiv 3 \pmod{4}$. Fix $k \in \mathbb{F}_p$, and for $1 \leq j \leq l = (p-1)/2$, define α_j by

$$\alpha_j = \begin{cases} kj & \text{if } \left(\frac{j}{p}\right) = 1 \\ -(k-1)j & \text{if } \left(\frac{j}{p}\right) = -1. \end{cases}$$

Then, $M(a, b)$ determined by the list $\{\alpha_1, \alpha_2, \dots, \alpha_l\}$ can be computed as

$$M(a, b) = \begin{cases} a & \text{if } a = b \\ ka - (k-1)b & \text{if } \left(\frac{a-b}{p}\right) = 1 \\ kb - (k-1)a & \text{if } \left(\frac{a-b}{p}\right) = -1. \end{cases}$$

Proof. We write $x \bmod p$ as $[x]_p$.

When $[a-b]_p \leq l$, if $\left(\frac{a-b}{p}\right) = 1$, then $M(a, b) = M(a-b, 0) + b = k(a-b) + b = ka - (k-1)b$.

if $\left(\frac{a-b}{p}\right) = -1$, then $M(a, b) = M(a-b, 0) + b = -(k-1)(a-b) + b = kb - (k-1)a$.

When $[b-a]_p \leq l$, it follows from (2) that

$$\left(\frac{b-a}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{a-b}{p}\right) = -\left(\frac{a-b}{p}\right).$$

If $\left(\frac{b-a}{p}\right) = 1$, then $M(a, b) = M(0, b-a) + a = k(b-a) + a = kb - (k-1)a$. If $\left(\frac{b-a}{p}\right) = -1$, then $M(a, b) = M(0, b-a) + a = -(k-1)(b-a) + a = ka - (k-1)b$. $\square$

4 Me Operation and Related Concepts

This section introduces the Me operation, defines Me-scalar multiplication and presents its properties, and discusses the hardness of the MeDLP, which is the Me version of the ECDLP, on classical computers.

In this section, we consider only primes p satisfying $p \equiv 3 \pmod{4}$. Note that (2) then holds.

Proposition 3 and Theorem 4(a) and (b) presented in this section were previously presented in [7], but only outlines of their proofs were given. Therefore, more detailed proofs are provided in this paper.

4.1 Me Operation

Let $E/\mathbb{F}_p$ be an elliptic curve given in the short Weierstrass form (1), and assume that $\#E(\mathbb{F}_p)$ is odd. Since

$$\exists x_0 \in \mathbb{F}_p, (x_0, 0) \in E(\mathbb{F}_p) \Leftrightarrow \#E(\mathbb{F}_p) \text{ is even,}$$

there is no point of the form $(x_0, 0) \in E(\mathbb{F}_p)$ when $\#E(\mathbb{F}_p)$ is odd. For $P = (x_0, y_0) \in E(\mathbb{F}_p)$, we have $-P = (x_0, -y_0) \in E(\mathbb{F}_p)$, and by (2),

$$\left\{ \begin{array}{l} \left(\frac{y_0}{p}\right) = 1, \\ \left(\frac{-y_0}{p}\right) = -1, \end{array} \right. \quad \text{or} \quad \left\{ \begin{array}{l} \left(\frac{y_0}{p}\right) = -1, \\ \left(\frac{-y_0}{p}\right) = 1, \end{array} \right. \quad (3)$$

holds.

Define the function $\text{sign} : E(\mathbb{F}_p) \setminus \{\mathcal{O}\} \rightarrow \{\pm 1\}$ by

$$\text{sign}((x_0, y_0)) = \left(\frac{y_0}{p} \right). \quad (4)$$

Then, by (3), for $P \in E(\mathbb{F}_p) \setminus \{\mathcal{O}\}$,

$$\text{sign}(P) \cdot \text{sign}(-P) = -1 \quad (5)$$

holds¹.

Based on the definition of the list-free M operation with $k = 2$ ², define the Me operation $\text{Me} : E(\mathbb{F}_p) \times E(\mathbb{F}_p) \rightarrow E(\mathbb{F}_p)$ by

$$\text{Me}(P, Q) = \begin{cases} P & \text{if } P = Q, \\ 2P - Q & \text{if } \text{sign}(P - Q) = 1, \\ 2Q - P & \text{if } \text{sign}(P - Q) = -1, \end{cases} \quad (6)$$

and denote it by

$$P \oplus Q := \text{Me}(P, Q).$$

The Me operation has the following properties.

Proposition 3 ([5], Proposition 12).

For $P, Q, R \in E(\mathbb{F}_p)$, the following properties hold.

- (a) $P \oplus P = P$. (*Idempotent law*)
- (b) $P \oplus Q = Q \oplus P$. (*Commutative law*)
- (c) $(P \oplus Q) + R = (P + R) \oplus (Q + R)$. (*Distributive law*)
- (d) In general, $(P \oplus Q) \oplus R \neq P \oplus (Q \oplus R)$. (*Nonassociativity*)

Proof. (a) This is trivial from the definition of the Me operation.

(b) The case $P = Q$ is obvious. Suppose $P \neq Q$. If $\text{sign}(P - Q) = 1$, then $P \oplus Q = 2P - Q$. Moreover, by (5), $\text{sign}(Q - P) = -1$, and hence $Q \oplus P = 2P - Q$. Similarly, if $\text{sign}(P - Q) = -1$, then $P \oplus Q = Q \oplus P = 2Q - P$.

(c) The case $P = Q$ follows immediately from (a). In the following, suppose $P \neq Q$.

If $\text{sign}(P - Q) = 1$, the left-hand side is $(P \oplus Q) + R = 2P - Q + R$. Since $\text{sign}((P + R) - (Q + R)) = \text{sign}(P - Q) = 1$, the right-hand side is $(P + R) \oplus (Q + R) = 2(P + R) - (Q + R) = 2P - Q + R$. If $\text{sign}(P - Q) = -1$, the left-hand side is $(P \oplus Q) + R = 2Q - P + R$, and the right-hand side is $(P + R) \oplus (Q + R) = 2(Q + R) - (P + R) = 2Q - P + R$. $\square$

4.2 Me-Scalar Multiplication

We define Me-scalar multiplication using the Me operation, corresponding to scalar multiplication of points on elliptic curves. There are two types of Me-scalar multiplication: 1st Me-scalar multiplication and 2nd Me-scalar multiplication. The 2nd Me-scalar multiplication has more properties, and its scalar can be extended to positive rational numbers.

4.2.1 1st Me-Scalar Multiplication

By Proposition 3(a), for $P \in E(\mathbb{F}_p)$,

$$(\cdots (((P \oplus P) \oplus P) \oplus P) \oplus \cdots P) \oplus P = P.$$

¹As long as (5) holds, the definition of the function $\text{sign} : E(\mathbb{F}_p) \rightarrow \{\pm 1\}$ may differ from (6) in the following discussion.

²We restrict k to 2 for the efficiency of computing the Me operation.

Hence, simple repeated applications of $\oplus$ are meaningless. We thus define the 1st Me-scalar multiplication $P_{n,Z}^I$ of P with auxiliary element $Z \in E(\mathbb{F}_p)$ as the output of Algorithm 1. Note that when $n = 1$, Steps 2 through 5 are not executed, and $P_{1,Z}^I = P$. The procedure of Algorithm 1 is the same as that of the ordinary left-to-right binary method if $Y \leftarrow (Z + Y) \oplus Y$ in Step 3 is replaced by $Y \leftarrow Y \oplus Y$.

Algorithm 1 (Computation of 1st Me-Scalar Multiplication)

Input: $P, Z \in E(\mathbb{F}_p), n = (1, n_{t-2}, \dots, n_0)_2 \in \mathbb{N}$

Output: $P_{n,Z}^I \in E(\mathbb{F}_p)$

1. $Y \leftarrow P$
 2. **for** $i = t - 2$ **down to** 0
 3. $Y \leftarrow (Z + Y) \oplus Y$
 4. **if** $n_i = 1$ **then** $Y \leftarrow Y \oplus P$
 5. **end for**
 6. **return** Y
-

It may be more natural to define the 1st Me-scalar multiplication by

$$P_{n,Z}^I = (\dots(((Z + P) \oplus P) \oplus P) \oplus \dots \oplus P) \quad (7)$$

However, under such a definition, the binary method cannot be applied to the computation of (7) because $\oplus$ does not satisfy the associative law. This is because the binary method is an algorithm that changes the order of operations by using the associative law. Therefore, under such a definition, computing the 1st Me-scalar multiplication $P_{n,Z}^I$ requires $n - 1$ computations of $\oplus$, and when n is large (e.g., several hundred bits), this computation is practically infeasible.

The 1st Me-scalar multiplication have the following properties.

Theorem 4 ([5], Theorem 14, Corollary 15).

Let $P, Q, Z \in E(\mathbb{F}_p)$ and $n, m \in \mathbb{N}$. Then the following hold.

- (a) $P_{n,Z}^I + Q = P + Q_{n,Z}^I = (P + Q)_{n,Z}^I$.
- (b) $(P_{n,Z}^I)_{m,Z}^I = (P_{m,Z}^I)_{n,Z}^I = P_{n,Z}^I + P_{m,Z}^I - P$. In general, $P_{nm,Z}^I \neq (P_{n,Z}^I)_{m,Z}^I$.
- (c) $P_{n,Z}^I \oplus Q_{n,Z}^I = (P \oplus Q)_{n,Z}^I$.

The following two lemmas are needed in the proof of Theorem 4.

Lemma 5 ([5], Theorem 14).

For $P, Q, Z \in E(\mathbb{F}_p)$, $(Z + (P \oplus Q)) \oplus (P \oplus Q) = ((Z + P) \oplus P) \oplus ((Z + Q) \oplus Q)$ holds.

Proof. We obtain the following calculations, which show that the left-hand side is equal to the right-hand side.

$$\begin{aligned}
\text{LHS} &= (Z + (P \oplus Q)) \oplus (P \oplus Q) \\
&= ((Z + P) \oplus (Z + Q)) \oplus (P \oplus Q) && \text{by Proposition 3(c)} \\
&= \text{Me}(\text{Me}(Z + P, Z + Q), \text{Me}(P, Q)) \\
&= \text{Me}(\text{Me}(P - Q, \mathcal{O}) + Z + Q, \text{Me}(P - Q, \mathcal{O}) + Q) && \text{by Proposition 3(c)} \\
&= \text{Me}((\text{Me}(P - Q, \mathcal{O}) + Z) + Q, \text{Me}(P - Q, \mathcal{O}) + Q) \\
&= \text{Me}(\text{Me}(P - Q, \mathcal{O}) + Z, \text{Me}(P - Q, \mathcal{O})) + Q && \text{by Proposition 3(c)} \\
&= \text{Me}(Z, \mathcal{O}) + \text{Me}(P - Q, \mathcal{O}) + Q, && \text{by Proposition 3(c)}
\end{aligned}$$

and

$$\begin{aligned}
\text{RHS} &= ((Z + P) \oplus P) \oplus ((Z + Q) \oplus Q) \\
&= \text{Me}(\text{Me}(Z + P, P), \text{Me}(Z + Q, Q)) \\
&= \text{Me}(\text{Me}(Z, \mathcal{O}) + P, \text{Me}(Z, \mathcal{O}) + Q) && \text{by Proposition 3(c)} \\
&= \text{Me}((\text{Me}(Z, \mathcal{O}) + P - Q) + Q, \text{Me}(Z, \mathcal{O}) + Q) \\
&= \text{Me}(\text{Me}(Z, \mathcal{O}) + P - Q, \text{Me}(Z, \mathcal{O})) + Q && \text{by Proposition 3(c)} \\
&= \text{Me}(P - Q, \mathcal{O}) + \text{Me}(Z, \mathcal{O}) + Q. && \text{by Proposition 3(c)}
\end{aligned}$$

□

Lemma 6 ([9], Lemma 7).

For $P, Z \in E(\mathbb{F}_p)$ and $n \in \mathbb{N}$,

$$P_{2n,Z}^I = (Z + P_{n,Z}^I) \oplus P_{n,Z}^I.$$

Proof. Let the binary representation of n be $(1, n_{l-2}, n_{l-3}, \dots, n_0)_2$. Then the binary representation of $2n$ is $(1, n_{l-2}, n_{l-3}, \dots, n_0, 0)_2$. When $P_{2n,Z}^I$ is computed using Algorithm 1, the value of Y at the beginning of the last iteration is $P_{n,Z}^I$. Therefore, in Step 3, $Y \leftarrow (Z + P_{n,Z}^I) \oplus P_{n,Z}^I$ is computed. Since the condition of the if statement in Step 4 is false, Step 4 is not executed, and $(Z + P_{n,Z}^I) \oplus P_{n,Z}^I$ is returned in Step 6. Hence, $P_{2n,Z}^I = (Z + P_{n,Z}^I) \oplus P_{n,Z}^I$. □

Proof of Theorem 4.

(a) In the computation of $P_{n,Z}^I$ using Algorithm 1, let $Y[P_{n,Z}^I]_{j,s}$ denote the value of Y after Step s is executed for $i = j$. We consider the following equations:

$$\begin{aligned}
Y[P_{n,Z}^I]_{t-2,3} + Q &= Y[Q_{n,Z}^I]_{t-2,3} + P \\
&= Y[(P + Q)_{n,Z}^I]_{t-2,3},
\end{aligned} \tag{8}$$

$$\begin{aligned}
Y[P_{n,Z}^I]_{j,3} + Q &= Y[Q_{n,Z}^I]_{j,3} + P \\
&= Y[(P + Q)_{n,Z}^I]_{j,3},
\end{aligned} \tag{9}$$

$$\begin{aligned}
Y[P_{n,Z}^I]_{j,4} + Q &= Y[Q_{n,Z}^I]_{j,4} + P \\
&= Y[(P + Q)_{n,Z}^I]_{j,4},
\end{aligned} \tag{10}$$

and

$$\begin{aligned}
Y[P_{n,Z}^I]_{j-1,3} + Q &= Y[Q_{n,Z}^I]_{j-1,3} + P \\
&= Y[(P + Q)_{n,Z}^I]_{j-1,3}.
\end{aligned} \tag{11}$$

For (8)–(11), to prove (a), it suffices to show the following:

- (i) Equation (8) holds.
- (ii) If (9) holds, then (10) holds.
- (iii) If (10) holds, then (11) holds.
- (i) By direct computation, we have

$$\begin{aligned}
Y[P_{n,Z}^I]_{t-2,3} + Q &= ((Z + P) \oplus P) + Q \\
&= ((Z + P) + Q) \oplus (P + Q) && \text{by Proposition 3(c)} \\
&= (Z + (P + Q)) \oplus (P + Q),
\end{aligned}$$

$$\begin{aligned}
Y[Q_{n,Z}^I]_{t-2,3} + P &= ((Z + Q) \oplus Q) + P \\
&= ((Z + Q) + P) \oplus (Q + P) && \text{by Proposition 3(c)} \\
&= (Z + (P + Q)) \oplus (P + Q),
\end{aligned}$$

and

$$Y[(P + Q)_{n,Z}^I]_{t-2,3} = (Z + (P + Q)) \oplus (P + Q).$$

Thus, (8) holds.

(ii) If $n_i = 0$, Step 4 of Algorithm 1 does nothing, and hence (9) and (10) are identical. Thus, (ii) holds.

If $n_i = 1$, let A denote the value in (9). Then, we have

$$\begin{aligned} Y[P_{n,Z}^I]_{j,3} &= A - Q, \\ Y[Q_{n,Z}^I]_{j,3} &= A - P, \\ Y[(P + Q)_{n,Z}^I]_{j,3} &= A. \end{aligned}$$

Therefore, we have

$$\begin{aligned} Y[P_{n,Z}^I]_{j,4} + Q &= ((A - Q) \oplus P) + Q \\ &= ((A - Q) + Q) \oplus (P + Q) \quad \text{by Proposition 3(c)} \\ &= A \oplus (P + Q), \\ Y[Q_{n,Z}^I]_{j,4} + P &= ((A - P) \oplus Q) + P \\ &= ((A - P) + P) \oplus (Q + P) \quad \text{by Proposition 3(c)} \\ &= A \oplus (P + Q), \end{aligned}$$

and

$$Y[(P + Q)_{n,Z}^I]_{j,4} = A \oplus (P + Q).$$

Hence, (ii) holds.

(iii) Let B denote the value in (10). Then we have

$$\begin{aligned} Y[P_{n,Z}^I]_{j,4} &= B - Q, \\ Y[Q_{n,Z}^I]_{j,4} &= B - P, \\ Y[(P + Q)_{n,Z}^I]_{j,4} &= B. \end{aligned}$$

Therefore, we have

$$\begin{aligned} Y[P_{n,Z}^I]_{j-1,3} + Q &= (Z + B - Q) \oplus (B - Q) + Q \\ &= (Z + B - Q + Q) \oplus (B - Q + Q) \quad \text{by Proposition 3(c)} \\ &= (Z + B) \oplus B, \\ Y[Q_{n,Z}^I]_{j-1,3} + P &= (Z + B - P) \oplus (B - P) + P \\ &= (Z + B - P + P) \oplus (B - P + P) \quad \text{by Proposition 3(c)} \\ &= (Z + B) \oplus B, \end{aligned}$$

and

$$Y[(P + Q)_{n,Z}^I]_{j-1,3} = (Z + B) \oplus B.$$

Hence, (iii) holds.

(b) Using (a), we have

$$\begin{aligned} (P_{n,Z}^I)_{m,Z}^I + P &= (P_{n,Z}^I + P)_{m,Z}^I \\ &= (P + P_{n,Z}^I)_{m,Z}^I \\ &= P_{m,Z}^I + P_{n,Z}^I \\ &= (P_{m,Z}^I)_{n,Z}^I + P. \end{aligned}$$

Hence, (b) holds.

(c) We prove (c) by induction. The case $n = 1$ is obvious.

Assume that

$$P_{n,Z}^I \oplus Q_{n,Z}^I = (P \oplus Q)_{n,Z}^I \quad (12)$$

holds for $n = 1, \dots, k-1$.

If $k \geq 2$ is even, (12) gives

$$P_{k/2,Z}^I \oplus Q_{k/2,Z}^I = (P \oplus Q)_{k/2,Z}^I. \quad (13)$$

Then, by Lemma 6,

$$P_{k,Z}^I \oplus Q_{k,Z}^I = ((Z + P_{k/2,Z}^I) \oplus P_{k/2,Z}^I) \oplus ((Z + Q_{k/2,Z}^I) \oplus Q_{k/2,Z}^I) \quad (14)$$

and by (13), we have

$$\begin{aligned} (P \oplus Q)_{k,Z}^I &= (Z + (P \oplus Q)_{k/2,Z}^I) \oplus (P \oplus Q)_{k/2,Z}^I \\ &= (Z + (P_{k/2,Z}^I \oplus Q_{k/2,Z}^I)) \oplus (P_{k/2,Z}^I \oplus Q_{k/2,Z}^I). \end{aligned} \quad (15)$$

By Lemma 5, (14) and (15) are equal.

If k is odd, (12) gives

$$P_{k-1,Z}^I \oplus Q_{k-1,Z}^I = (P \oplus Q)_{k-1,Z}^I.$$

By (a),

$$P_{k-1,Z}^I - P = Q_{k-1,Z}^I - Q.$$

Let this value be X . Then we have

$$\begin{cases} P_{k-1,Z}^I = X + P, \\ Q_{k-1,Z}^I = X + Q. \end{cases}$$

Therefore, by Algorithm 1 and Lemma 5, we have

$$\begin{aligned} P_{k,Z}^I \oplus Q_{k,Z}^I &= (P_{k-1,Z}^I \oplus P) \oplus (Q_{k-1,Z}^I \oplus Q) \\ &= ((X + P) \oplus P) \oplus ((X + Q) \oplus Q) \\ &= (X + (P \oplus Q)) \oplus (P \oplus Q). \end{aligned} \quad (16)$$

By (a),

$$\underbrace{P_{k-1,Z}^I - P}_{=X} + (P \oplus Q) = (P \oplus Q)_{k-1,Z}^I.$$

Therefore, by (16),

$$\begin{aligned} P_{k,Z}^I \oplus Q_{k,Z}^I &= (X + (P \oplus Q)) \oplus (P \oplus Q) \\ &= (P \oplus Q)_{k-1,Z}^I \oplus (P \oplus Q) \\ &= (P \oplus Q)_{k,Z}^I. \end{aligned}$$

Thus, (c) holds. $\square$

4.2.2 2nd Me-Scalar Multiplication

Let $P, Z \in E(\mathbb{F}_p)$ and $n \in \mathbb{N}$. We define the 2nd Me-scalar multiplication $P_{n,Z}^{II}$ of P with auxiliary element Z as follows.

- If $n = 1$, define $P_{1,Z}^{II} = P_{1,Z}^I = P$.

- If $n \geq 2$, let $n = p_1 p_2 \cdots p_k$ be the prime factorization of n , where repeated prime factors are allowed. Define

$$P_{n,Z}^{II} = (\cdots ((P_{p_1,Z}^I)^I_{p_2,Z}) \cdots)^I_{p_k,Z}. \quad (17)$$

For example, since $20 = 2 \cdot 2 \cdot 5$,

$$P_{20,Z}^{II} = ((P_{2,Z}^I)^I_{2,Z})^I_{5,Z}.$$

The computation in (17) can be written as Algorithm 2.

Algorithm 2 (Computation of 2nd Me-Scalar Multiplication)	
Input:	$P, Z \in E(\mathbb{F}_p), n = p_1 p_2 \cdots p_k$
Output:	$P_{n,Z}^{II} \in E(\mathbb{F}_p)$
1.	$Y \leftarrow P$
2.	for $i = 1$ to k
3.	$Y \leftarrow Y_{p_i,Z}^I$
4.	end for
5.	return Y

By Theorem 4(b), (17) can be rewritten as

$$P_{n,Z}^{II} = P_{p_1,Z}^I + P_{p_2,Z}^I + \cdots + P_{p_k,Z}^I - (k-1)P. \quad (18)$$

This means that the computation in (17) is independent of the order of $p_1, p_2, \dots, p_k$. To compute $P_{n,Z}^{II}$, n must be factored into primes.

Proposition 7 ([11], Proposition 3).

Let $P, Q, Z \in E(\mathbb{F}_p)$ and $n, m \in \mathbb{N}$. Then the following hold.

- (a) $P_{n,Z}^{II} + Q = P + Q_{n,Z}^{II} = (P + Q)_{n,Z}^{II}$.
- (b) $(P_{n,Z}^{II})_{m,Z}^{II} = (P_{m,Z}^{II})_{n,Z}^{II} = P_{n,Z}^{II} + P_{m,Z}^{II} - P = P_{nm,Z}^{II}$.
- (c) $P_{n,Z}^{II} \oplus Q_{n,Z}^{II} = (P \oplus Q)_{n,Z}^{II}$.

Proof. The proposition is obvious when $n = 1$. In the following, assume that $n \geq 2$.

- (a) We show (i) $P_{n,Z}^{II} + Q = P + Q_{n,Z}^{II}$ and (ii) $P_{n,Z}^{II} + Q = (P + Q)_{n,Z}^{II}$.

For (i), let $n = p_1 p_2 \cdots p_k$. Then we have

$$\begin{aligned}
P_{n,Z}^{II} + Q &= P_{p_1,Z}^I + P_{p_2,Z}^I + \cdots + P_{p_k,Z}^I - (k-1)P + Q && \text{by (18)} \\
&= (P_{p_1,Z}^I + Q) + (P_{p_2,Z}^I + Q) + \cdots + (P_{p_k,Z}^I + Q) \\
&\quad - (k-1)P - (k-1)Q \\
&= (P + Q_{p_1,Z}^I) + (P + Q_{p_2,Z}^I) + \cdots + (P + Q_{p_k,Z}^I) \\
&\quad - (k-1)P - (k-1)Q && \text{by Theorem 4(a)} \\
&= P + (Q_{p_1,Z}^I + Q_{p_2,Z}^I + \cdots + Q_{p_k,Z}^I - (k-1)Q) \\
&= P + Q_{n,Z}^{II}.
\end{aligned}$$

For (ii), for $R \in E(\mathbb{F}_p)$, we have

$$\begin{aligned}
(P_{n,Z}^{II} + Q) + R &= (P_{n,Z}^{II} + R) + Q \\
&= (P + R_{n,Z}^{II}) + Q && \text{by (i)} \\
&= (P + Q) + R_{n,Z}^{II} \\
&= (P + Q)_{n,Z}^{II} + R. && \text{by (i)}
\end{aligned}$$

(b) Let $n = p_1 p_2 \cdots p_k$ and $m = q_1 q_2 \cdots q_{k'}$. For $(P_{n,Z}^{II})_{m,Z}^{II}$, we have

$$\begin{aligned}
(P_{n,Z}^{II})_{m,Z}^{II} &= ((P_{n,Z}^{II})_{m,Z}^{II} + P) - P \\
&= (P_{n,Z}^{II} + P_{m,Z}^{II}) - P && \text{by (a)} \\
&= (P_{p_1,Z}^I + P_{p_2,Z}^I + \cdots + P_{p_k,Z}^I - (k-1)P) \\
&\quad + (P_{q_1,Z}^I + P_{q_2,Z}^I + \cdots + P_{q_{k'},Z}^I - (k'-1)P) - P && \text{by (18)} \\
&= P_{p_1,Z}^I + \cdots + P_{p_k,Z}^I + P_{q_1,Z}^I + \cdots + P_{q_{k'},Z}^I \\
&\quad - (k + k' - 1)P.
\end{aligned}$$

The above calculation also shows that $(P_{n,Z}^{II})_{m,Z}^{II} = P_{n,Z}^{II} + P_{m,Z}^{II} - P$. Similarly, for $(P_{m,Z}^{II})_{n,Z}^{II}$, we have

$$(P_{m,Z}^{II})_{n,Z}^{II} = P_{p_1,Z}^I + \cdots + P_{p_k,Z}^I + P_{q_1,Z}^I + \cdots + P_{q_{k'},Z}^I - (k + k' - 1)P.$$

Since $nm = p_1 p_2 \cdots p_k q_1 q_2 \cdots q_{k'}$ is the prime factorization of nm , by (18), we have

$$P_{nm,Z}^{II} = P_{p_1,Z}^I + \cdots + P_{p_k,Z}^I + P_{q_1,Z}^I + \cdots + P_{q_{k'},Z}^I - (k + k' - 1)P.$$

Hence, (b) holds.

(c) We have

$$\begin{aligned}
P_{n,Z}^{II} \oplus Q_{n,Z}^{II} &= (\mathcal{O} + P_{n,Z}^{II}) \oplus (\mathcal{O} + Q_{n,Z}^{II}) \\
&= (\mathcal{O}_{n,Z}^{II} + P) \oplus (\mathcal{O}_{n,Z}^{II} + Q) && \text{by (a)} \\
&= \mathcal{O}_{n,Z}^{II} + (P \oplus Q) && \text{by Proposition 3(c)} \\
&= \mathcal{O} + (P \oplus Q)_{n,Z}^{II} && \text{by (a)} \\
&= (P \oplus Q)_{n,Z}^{II}.
\end{aligned}$$

Thus, (c) holds. □

Proposition 8 ([12], Proposition 20).

For $P, Z \in E(\mathbb{F}_p)$ and a prime n ,

$$P_{n,Z}^{II} = P_{n,Z}^I.$$

Proof. This is trivial from the definition of the 2nd Me-scalar multiplication. □

The scalar of the 2nd Me-scalar multiplication can be extended to the positive rational numbers $\mathbb{Q}_+$ by the following Definition 9.

Definition 9.

For $P, Z \in E(\mathbb{F}_p)$ and $n = n_0/n_1 \in \mathbb{Q}_+$ ($n_0, n_1 \in \mathbb{N}$), define

$$P_{n_0/n_1,Z}^{II} := P_{n_0,Z}^{II} - P_{n_1,Z}^{II} + P.$$

By the following proposition, this definition is well-defined.

Proposition 10 ([11], Proposition 6).

Let $P, Z \in E(\mathbb{F}_p)$ and $n_0, n_1, m_0, m_1 \in \mathbb{N}$. If $n_0/n_1 = m_0/m_1$ as elements of $\mathbb{Q}_+$, then

$$P_{n_0/n_1,Z}^{II} = P_{m_0/m_1,Z}^{II}.$$

Proof. By the assumption, $n_0 m_1 = m_0 n_1$. Hence, we have

$$\begin{aligned}
P_{n_0 m_1, Z}^{II} &= P_{m_0 n_1, Z}^{II}, \\
P_{n_0 m_1, Z}^{II} + P &= P_{m_0 n_1, Z}^{II} + P, \\
P_{n_0, Z}^{II} + P_{m_1, Z}^{II} &= P_{m_0, Z}^{II} + P_{n_1, Z}^{II}, & \text{by Proposition 7(b)} \\
P_{n_0, Z}^{II} - P_{n_1, Z}^{II} &= P_{m_0, Z}^{II} - P_{m_1, Z}^{II}, \\
\underbrace{P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P}_{=P_{n_0/n_1, Z}^{II}} &= \underbrace{P_{m_0, Z}^{II} - P_{m_1, Z}^{II} + P}_{=P_{m_0/m_1, Z}^{II}}.
\end{aligned}$$

□

Remark 11.

To extend the scalar of the 1st Me-scalar multiplication to $\mathbb{Q}_+$, it is necessary that

$$P_{nm, Z}^I = P_{n, Z}^I + P_{m, Z}^I - P$$

hold for $n, m \in \mathbb{N}$. However, this does not hold in general.

Theorem 12 ([11], Theorem 7).

Let $P, Q, Z \in E(\mathbb{F}_p)$ and $n, m \in \mathbb{Q}_+$. Then the following hold.

- (a) $P_{n, Z}^{II} + Q = P + Q_{n, Z}^{II} = (P + Q)_{n, Z}^{II}$.
- (b) $(P_{n, Z}^{II})_{m, Z}^{II} = (P_{m, Z}^{II})_{n, Z}^{II} = P_{n, Z}^{II} + P_{m, Z}^{II} - P = P_{nm, Z}^{II}$.
- (c) $P_{n, Z}^{II} \oplus Q_{n, Z}^{II} = (P \oplus Q)_{n, Z}^{II}$.

Proof. Let $n = n_0/n_1, m = m_0/m_1, n_0, n_1, m_0, m_1 \in \mathbb{N}$.

(a) By the following calculations, (a) holds:

$$\begin{aligned}
P_{n_0/n_1, Z}^{II} + Q &= P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P + Q, \\
P + Q_{n_0/n_1, Z}^{II} &= P + Q_{n_0, Z}^{II} - Q_{n_1, Z}^{II} + Q \\
&= P + (P + Q_{n_0, Z}^{II}) - (P + Q_{n_1, Z}^{II}) + Q \\
&= P + (P_{n_0, Z}^{II} + Q) - (P_{n_1, Z}^{II} + Q) + Q & \text{by Proposition 7(a)} \\
&= P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P + Q, \\
(P + Q)_{n_0/n_1, Z}^{II} &= (P + Q)_{n_0, Z}^{II} - (P + Q)_{n_1, Z}^{II} + (P + Q) \\
&= (P_{n_0, Z}^{II} + Q) - (P_{n_1, Z}^{II} + Q) + (P + Q) & \text{by Proposition 7(a)} \\
&= P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P + Q.
\end{aligned}$$

(b) By direct computation, $(P_{n, Z}^{II})_{m, Z}^{II}$ is given as follows:

$$\begin{aligned}
(P_{n, Z}^{II})_{m, Z}^{II} &= (P_{n, Z}^{II})_{m_0/m_1, Z}^{II} \\
&= (P_{n, Z}^{II})_{m_0, Z}^{II} - (P_{n, Z}^{II})_{m_1, Z}^{II} + P_{n, Z}^{II} \\
&= (P_{n_0/n_1, Z}^{II})_{m_0, Z}^{II} - (P_{n_0/n_1, Z}^{II})_{m_1, Z}^{II} + P_{n_0/n_1, Z}^{II} \\
&= (P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P)_{m_0, Z}^{II} - (P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P)_{m_1, Z}^{II} + (P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P) & \text{by (a)} \\
&= (P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P_{m_0, Z}^{II}) - (P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P_{m_1, Z}^{II}) + (P_{n_0, Z}^{II} - P_{n_1, Z}^{II} + P) \\
&= P_{n_0, Z}^{II} + P_{m_0, Z}^{II} - P_{n_1, Z}^{II} - P_{m_1, Z}^{II} + P
\end{aligned}$$

Similarly, we have $(P_{m,Z}^{II})_{n,Z}^{II} = P_{n_0,Z}^{II} + P_{m_0,Z}^{II} - P_{n_1,Z}^{II} - P_{m_1,Z}^{II} + P$.

For $P_{n,Z}^{II} + P_{m,Z}^{II} - P$ and $P_{nm,Z}^{II}$, we have

$$\begin{aligned}
P_{n,Z}^{II} + P_{m,Z}^{II} - P &= P_{n_0/n_1,Z}^{II} + P_{m_0/m_1,Z}^{II} - P \\
&= (P_{n_0,Z}^{II} - P_{n_1,Z}^{II} + P) + (P_{m_0,Z}^{II} - P_{m_1,Z}^{II} + P) - P \\
&= P_{n_0,Z}^{II} + P_{m_0,Z}^{II} - P_{n_1,Z}^{II} - P_{m_1,Z}^{II} + P, \\
P_{nm,Z}^{II} &= P_{(n_0/n_1) \cdot (m_0/m_1),Z}^{II} \\
&= P_{(n_0 m_0)/(n_1 m_1),Z}^{II} \\
&= P_{n_0 m_0,Z}^{II} - P_{n_1 m_1,Z}^{II} + P \\
&= (P_{n_0,Z}^{II} + P_{m_0,Z}^{II} - P) - (P_{n_1,Z}^{II} + P_{m_1,Z}^{II} - P) + P \quad \text{by Proposition 7(b)} \\
&= P_{n_0,Z}^{II} + P_{m_0,Z}^{II} - P_{n_1,Z}^{II} - P_{m_1,Z}^{II} + P.
\end{aligned}$$

Hence, (b) holds.

(c) By the following calculation proves (c):

$$\begin{aligned}
\text{LHS} &= P_{n,Z}^{II} \oplus Q_{n,Z}^{II} \\
&= (\mathcal{O} + P_{n,Z}^{II}) \oplus (\mathcal{O} + Q_{n,Z}^{II}) \\
&= (\mathcal{O}_{n,Z}^{II} + P) \oplus (\mathcal{O}_{n,Z}^{II} + Q) && \text{by (a)} \\
&= \mathcal{O}_{n,Z}^{II} + (P \oplus Q) && \text{by Proposition 3(c)} \\
&= \mathcal{O} + (P \oplus Q)_{n,Z}^{II} && \text{by (a)} \\
&= (P \oplus Q)_{n,Z}^{II} \\
&= \text{RHS}.
\end{aligned}$$

□

4.3 MeDLP and MeCDH Problems

We define Me-scalar multiplication versions of the discrete logarithm problem and the CDH problem.

Definition 13.

(a) Given $P, Z, Q \in E(\mathbb{F}_p)$, the problem of finding an $n \in \mathbb{N}$ satisfying $Q = P_{n,Z}^I$ is called the discrete logarithm problem for the 1st Me scalar multiplication (1st MeDLP). We denote this problem by $\text{MeDLP}_I(P, Z, Q)$.

(b) For $P, Z \in E(\mathbb{F}_p)$ and $n, m \in \mathbb{N}$, let $Q_1 = P_{n,Z}^I$ and $Q_2 = P_{m,Z}^I$. The problem of computing $(P_{n,Z}^I)_{m,Z}^I$ from P, Z, Q_1 , and Q_2 is called the computational Diffie–Hellman problem for the 1st Me-scalar multiplication (1st MeCDH problem). We denote this problem by $\text{MeCDH}_I(P, Z, Q_1, Q_2)$.

Definition 14.

(a) Given $P, Z, Q \in E(\mathbb{F}_p)$, the problem of finding an $n \in \mathbb{Q}_+$ satisfying $Q = P_{n,Z}^{II}$ is called the discrete logarithm problem for the 2nd Me-scalar multiplication (2nd MeDLP). We denote this problem by $\text{MeDLP}_{II}(P, Z, Q)$.

(b) For $P, Z \in E(\mathbb{F}_p)$ and $n, m \in \mathbb{Q}_+$, let $Q_1 = P_{n,Z}^{II}$, $Q_2 = P_{m,Z}^{II}$. The problem of computing $P_{nm,Z}^{II}$ from P, Z, Q_1 , and Q_2 is called the computational Diffie–Hellman problem for the 2nd Me-scalar multiplication (2nd MeCDH problem). We denote this problem by $\text{MeCDH}_{II}(P, Z, Q_1, Q_2)$.

The following conditions are needed to discuss the hardness of the 1st MeDLP and the 2nd MeDLP.

Condition 15.

- (a) $E(\mathbb{F}_p)$ is cyclic, $\text{sign}(Z) = 1$, and $2Z$ is a generator of $E(\mathbb{F}_p)$.
- (a') $\#E(\mathbb{F}_p)$ is prime, $\text{sign}(Z) = 1$, and $2Z$ is a generator of $E(\mathbb{F}_p)$.
- (b) $E(\mathbb{F}_p)$ is cyclic, $\text{sign}(Z) = -1$, and $-Z$ is a generator of $E(\mathbb{F}_p)$.
- (b') $\#E(\mathbb{F}_p)$ is prime, $\text{sign}(Z) = -1$, and $-Z$ is a generator of $E(\mathbb{F}_p)$.
- (c) $\#E(\mathbb{F}_p)$ is a prime greater than or equal to 3, and $Z \neq \mathcal{O}$.

Condition 15(a') is stronger than Condition 15(a), and Condition 15(b') is stronger than Condition 15(b). If Condition 15(c) holds, then either Condition 15(a') or Condition 15(b') holds, and hence either Condition 15(a) or Condition 15(b) holds.

4.4 Existence of Solutions to the MeDLP

The following lemma is needed to prove the existence of solutions to the MeDLP.

Lemma 16 ([9], Lemma 8, [12], Lemma 23).

Let $X \in \{I, II\}$. For any $Z \in E(\mathbb{F}_p)$, $Z \neq \mathcal{O}$ and $n \in \mathbb{N}$, we have

$$\mathcal{O}_{2^n, Z}^X = \begin{cases} 2nZ & \text{if } \text{sign}(Z) = 1, \\ -nZ & \text{if } \text{sign}(Z) = -1. \end{cases}$$

Proof. We first consider the case $X = I$.

Suppose that $\text{sign}(Z) = 1$. We prove by induction that

$$\mathcal{O}_{2^n, Z}^I = 2nZ. \tag{19}$$

When $n = 1$, by the definition of the 1st Me-scalar multiplication,

$$\mathcal{O}_{2, Z}^I = (Z + \mathcal{O}) \oplus \mathcal{O} = Z \oplus \mathcal{O}.$$

Since

$$\text{sign}(Z - \mathcal{O}) = \text{sign}(Z) = 1,$$

by setting $P = Z$ and $Q = \mathcal{O}$ in (6), we have

$$Z \oplus \mathcal{O} = 2Z - \mathcal{O} = 2Z.$$

Assume that (19) holds for $n = k$, that is,

$$\mathcal{O}_{2^k, Z}^I = 2kZ. \tag{20}$$

Then,

$$\begin{aligned} \mathcal{O}_{2^{k+1}, Z}^I &= (Z + \mathcal{O}_{2^k, Z}^I) \oplus \mathcal{O}_{2^k, Z}^I && \text{by Lemma 6} \\ &= (Z + 2kZ) \oplus 2kZ && \text{by (20)} \\ &= (2k + 1)Z \oplus 2kZ. \end{aligned}$$

Moreover,

$$\text{sign}((2k + 1)Z - 2kZ) = \text{sign}(Z) = 1.$$

Therefore, by setting $P = (2k + 1)Z$ and $Q = 2kZ$ in (6), we have

$$(2k + 1)Z \oplus 2kZ = 2(2k + 1)Z - 2kZ = 2(k + 1)Z.$$

Hence, (19) holds for $n = k + 1$.

Suppose that $\text{sign}(Z) = -1$. We prove by induction that

$$\mathcal{O}_{2^n, Z}^I = -nZ. \quad (21)$$

When $n = 1$, by the definition of the 1st Me-scalar multiplication,

$$\mathcal{O}_{2, Z}^I = (Z + \mathcal{O}) \oplus \mathcal{O} = Z \oplus \mathcal{O}.$$

Since

$$\text{sign}(Z - \mathcal{O}) = \text{sign}(Z) = -1$$

by setting $P = Z$ and $Q = \mathcal{O}$ in (6), we have

$$Z \oplus \mathcal{O} = 2\mathcal{O} - Z = -Z.$$

Assume that (21) holds for $n = k$, that is,

$$\mathcal{O}_{2^k, Z}^I = -kZ. \quad (22)$$

Then,

$$\begin{aligned} \mathcal{O}_{2^{k+1}, Z}^I &= (Z + \mathcal{O}_{2^k, Z}^I) \oplus \mathcal{O}_{2^k, Z}^I && \text{by Lemma 6} \\ &= (Z - kZ) \oplus (-kZ) && \text{by (22)} \\ &= -(k-1)Z \oplus (-kZ). \end{aligned}$$

Moreover,

$$\text{sign}(-(k-1)Z - (-kZ)) = \text{sign}(Z) = -1.$$

Therefore, by setting $P = -(k-1)Z$ and $Q = -kZ$ in (6), we have

$$(-(k-1)Z) \oplus (-kZ) = 2(-kZ) - (-(k+1)Z) = (-k-1)Z.$$

Hence, (21) holds for $n = k+1$. Thus, the case $X = I$ has been proved.

Next, consider the case $X = II$. We have

$$\begin{aligned} \mathcal{O}_{2^n, Z}^{II} &= \underbrace{\mathcal{O}_{2, Z}^I + \cdots + \mathcal{O}_{2, Z}^I}_{n \text{ terms}} - (n-1)\mathcal{O} \quad \text{by (18)} \\ &= n\mathcal{O}_{2, Z}^I. \end{aligned}$$

By (19) and (21),

$$\mathcal{O}_{2, Z}^I = \begin{cases} 2Z & \text{if } \text{sign}(Z) = 1, \\ -Z & \text{if } \text{sign}(Z) = -1. \end{cases}$$

Therefore, the lemma also holds for $X = II$. □

Proposition 17 ([9], Lemma 9, Corollary 11, [12], Proposition 24).

Let $X \in \{I, II\}$. Suppose that $E(\mathbb{F}_p)$ and $Z \in E(\mathbb{F}_p)$ satisfy one of Conditions 15(a), (b), and (c). Then, for any $P \in E(\mathbb{F}_p)$,

$$\{P_{n, Z}^X : n = 1, 2, 3, \dots\} = E(\mathbb{F}_p).$$

Proof. For any $P \in E(\mathbb{F}_p)$, it is clear that

$$\{P_{n, Z}^X : n = 1, 2, 3, \dots\} \subset E(\mathbb{F}_p). \quad (23)$$

Suppose that $Z \in E(\mathbb{F}_p)$ satisfies Condition 15(a). Then we have

$$\begin{aligned}
E(\mathbb{F}_p) &= \{2nZ : n = 1, 2, 3, \dots\} && \text{since } 2Z \text{ is a generator of } E(\mathbb{F}_p) \\
&= \{\mathcal{O}_{2^n, Z}^X : n = 1, 2, 3, \dots\} && \text{by Lemma 16} \\
&= \{\mathcal{O}_{2^n, Z}^X + P : n = 1, 2, 3, \dots\} && \text{since } (E(\mathbb{F}_p), +) \text{ is a group} \\
&= \{\mathcal{O} + P_{2^n, Z}^X : n = 1, 2, 3, \dots\} && \text{by Theorem 4(a) or Proposition 7(a)} \\
&= \{P_{2^n, Z}^X : n = 1, 2, 3, \dots\}
\end{aligned}$$

Since $\{2^n : n = 1, 2, 3, \dots\} \subset \{n : n = 1, 2, 3, \dots\}$, we have

$$\{P_{2^n, Z}^X : n = 1, 2, 3, \dots\} \subset \{P_{n, Z}^X : n = 1, 2, 3, \dots\}.$$

Thus,

$$E(\mathbb{F}_p) \subset \{P_{n, Z}^X : n = 1, 2, 3, \dots\}. \quad (24)$$

By (23) and (24), we obtain

$$\{P_{n, Z}^X : n = 1, 2, 3, \dots\} = E(\mathbb{F}_p).$$

Similarly, when $Z \in E(\mathbb{F}_p)$ satisfies Condition 15(b), we can show that

$$\{P_{n, Z}^X : n = 1, 2, 3, \dots\} = E(\mathbb{F}_p).$$

If Condition 15(c) holds, then either Condition 15(a) or Condition 15(b) holds. Hence, the proposition also holds in this case. $\square$

The following two propositions show the existence of solutions to the 1st MeDLP and the 2nd MeDLP.

Proposition 18 ([9], Theorem 10).

Suppose that $E(\mathbb{F}_p)$ and $Z \in E(\mathbb{F}_p)$ satisfy one of Conditions 15(a), (b), and (c). Then, for any $P, Q \in E(\mathbb{F}_p)$, $\text{MeDLP}_I(P, Z, Q)$ has a solution $n \in \mathbb{N}$.

Proof. By Proposition 17, $Q \in E(\mathbb{F}_p) = \{P_{n, Z}^I : n = 1, 2, 3, \dots\}$. Hence, for some $n' \in \mathbb{N}$, $Q = P_{n', Z}^I$. This n' is a solution to $\text{MeDLP}_I(P, Z, Q)$. $\square$

Proposition 19 ([12], Proposition 25).

Suppose that $E(\mathbb{F}_p)$ and $Z \in E(\mathbb{F}_p)$ satisfy one of Conditions 15(a), (b), and (c). Then, for any $P, Q \in E(\mathbb{F}_p)$, $\text{MeDLP}_{II}(P, Z, Q)$ has a solution $n \in \mathbb{Q}_+$.

Proof. By Proposition 17, $Q \in E(\mathbb{F}_p) = \{P_{n, Z}^{II} : n = 1, 2, 3, \dots\}$. Hence, for some $n' \in \mathbb{N} \subset \mathbb{Q}_+$, $Q = P_{n', Z}^{II}$. This n' is a solution to $\text{MeDLP}_{II}(P, Z, Q)$. $\square$

4.5 Hardness of the MeDLP

The purpose of this subsection is to show the following:

- 1) A solution to $\text{ECDLP}(P, Q)$ can be constructed from a solution n to $\text{MeDLP}_I(P, Z, Q)$.
- 2) A solution to $\text{ECDLP}(P, Q)$ can be constructed from a solution n_0/n_1 to $\text{MeDLP}_{II}(P, Z, Q)$, provided that n_0 and n_1 can be factored.

Here, the auxiliary element Z is of the form $Z = zP$ ($z \in \mathbb{Z}/l\mathbb{Z}$), where z is assumed to be known.

Algorithm 3 presented in this subsection has a structure similar to that of Algorithm 1 and is needed to perform 1). By Theorems 4(b) and 12(b), the solutions to $\text{MeCDH}_I(P, Z, Q_1, Q_2)$ and $\text{MeCDH}_{II}(P, Z, Q_1, Q_2)$ are both $Q_1 + Q_2 - P$, and hence these problems are not difficult at all.

4.5.1 Algorithm 3

Let $l = \#E(\mathbb{F}_p)$. For $\mathcal{P} \in E(\mathbb{F}_p) \times \mathbb{Z}/l\mathbb{Z}$, let $\mathcal{P}[1]$ and $\mathcal{P}[2]$ denote the first and second components of $\mathcal{P}$, respectively. We define the operation $+$ on $E(\mathbb{F}_p) \times \mathbb{Z}/l\mathbb{Z}$ as

$$\mathcal{P} + \mathcal{Q} = (\mathcal{P}[1] + \mathcal{Q}[1], \mathcal{P}[2] + \mathcal{Q}[2]).$$

Define the operation $\oplus$ on $E(\mathbb{F}_p) \times \mathbb{Z}/l\mathbb{Z}$ as follows:

$$\mathcal{P} \oplus \mathcal{Q} = \begin{cases} (\mathcal{P}[1], \mathcal{Q}[1]) & \text{if } \mathcal{P}[1] = \mathcal{Q}[1] \\ (2\mathcal{P}[1] - \mathcal{Q}[1], 2\mathcal{P}[2] - \mathcal{Q}[2]) & \text{if } \text{sign}(\mathcal{P}[1] - \mathcal{Q}[1]) = 1 \\ (2\mathcal{Q}[1] - \mathcal{P}[1], 2\mathcal{Q}[2] - \mathcal{P}[2]) & \text{if } \text{sign}(\mathcal{P}[1] - \mathcal{Q}[1]) = -1 \end{cases}$$

Considering only the first component, the operation $\oplus$ on $E(\mathbb{F}_p) \times \mathbb{Z}/l\mathbb{Z}$ is the same as the operation $\oplus$ on $E(\mathbb{F}_p)$. Therefore,

$$(\mathcal{P} \oplus \mathcal{Q})[1] = \mathcal{P}[1] \oplus \mathcal{Q}[1]. \quad (25)$$

Algorithm 3 takes an element of $E(\mathbb{F}_p) \times \mathbb{Z}/l\mathbb{Z}$ as input and has a structure similar to that of Algorithm 1.

Algorithm 3	
Input:	$P \in E(\mathbb{F}_p), z \in \mathbb{Z}/l\mathbb{Z}, n = (1, n_{t-2}, \dots, n_0)_2 \in \mathbb{N}$
Output:	$\mathcal{Y} \in E(\mathbb{F}_p) \times \mathbb{Z}/l\mathbb{Z}$
1.	$\mathcal{P} = (P, 1)$
2.	$\mathcal{Z} = (zP, z)$
3.	$\mathcal{Y} = \mathcal{P}$
4.	for $i = t - 2$ down to 0
5.	$\mathcal{Y} = (\mathcal{Z} + \mathcal{Y}) \oplus \mathcal{Y}$
6.	if $n_i = 1$ then $\mathcal{Y} = \mathcal{Y} \oplus \mathcal{P}$
7.	end for
8.	return $\mathcal{Y}$

Algorithm 3 has the following properties.

Proposition 20 ([8], Lemma 18).

Let $\mathcal{Y}'$ be the output of Algorithm 3 with inputs $P \in E(\mathbb{F}_p)$, $z \in \mathbb{Z}/l\mathbb{Z}$, and $n \in \mathbb{N}$. Then the following hold.

- (a) $\mathcal{Y}'[1] = P_{n,zP}^I$.
- (b) $\mathcal{Y}'[2]P = P_{n,zP}^I$.

Proof. (a) By (25), the processing of the first component in Algorithm 3 with inputs $P \in E(\mathbb{F}_p)$, $z \in \mathbb{Z}/l\mathbb{Z}$, and $n \in \mathbb{N}$ is identical to the processing in Algorithm 1 with inputs $P, Z = zP \in E(\mathbb{F}_p)$ and $n \in \mathbb{N}$. Therefore, (a) holds.

(b) The initial values $\mathcal{P}$ and $\mathcal{Z}$ satisfy $\mathcal{P}[1] = \mathcal{P}[2]P$ and $\mathcal{Z}[1] = \mathcal{Z}[2]P$. Moreover, from the definitions of $+$ and $\oplus$, these relations are preserved at every step of Algorithm 3. Therefore, at every step of Algorithm 3,

$$\mathcal{Y}[1] = \mathcal{Y}[2]P.$$

Hence,

$$\mathcal{Y}'[2]P = \mathcal{Y}'[1] = P_{n,zP}^I.$$

□

4.5.2 Hardness of the 1st MeDLP

This subsection gives a theorem on the hardness of the 1st MeDLP.

Theorem 21 ([8], Theorem 19).

For given $P, Q \in E(\mathbb{F}_p)$ and $z \in \mathbb{Z}$, suppose that $E(\mathbb{F}_p)$ and $Z = zP$ satisfy one of Conditions 15(a'), (b'), and (c).

- (a) A solution to $\text{ECDLP}(P, Q)$ can be computed in polynomial time from a solution to $\text{MeDLP}_I(P, zP, Q)$.
- (b) If there exists an algorithm $\mathcal{A}$ that solves $\text{MeDLP}_I(P, zP, Q)$ in polynomial time, then $\text{ECDLP}(P, Q)$ can be solved in polynomial time.

Proof. (a) Suppose that $\text{MeDLP}_I(P, zP, Q)$ can be solved and that an $n \in \mathbb{N}$ satisfying $Q = P_{n,zP}^I$ is obtained. By Proposition 18, such an n exists. Running Algorithm 3 with inputs (P, z, n) , by Proposition 20(b), the second component t of its output satisfies $tP = Q$. Thus, t is a solution to $\text{ECDLP}(P, Q)$.

(b) This is a restatement of (a). □

4.5.3 Hardness of the 2nd MeDLP

First, we give two lemmas needed to discuss the hardness of the 2nd MeDLP, and then discuss the hardness of the 2nd MeDLP.

Lemma 22 ([12], Lemma 26).

For $n \in \mathbb{N}$, let $Q = P_{n,zP}^{II}$. Let $n = p_1 p_2 \cdots p_k$ be a prime factorization of n . Let n_i be the second component of the output of Algorithm 3 with inputs (P, z, p_i) . Then,

$$Q = ((n_1 + n_2 + \cdots + n_k) - (k - 1))P.$$

Proof. By Propositions 8 and 20(b),

$$P_{p_i,zP}^{II} = P_{p_i,zP}^I = n_i P. \tag{26}$$

Therefore, we have

$$\begin{aligned} Q &= P_{n,zP}^{II} \\ &= P_{p_1,zP}^{II} + P_{p_2,zP}^{II} + \cdots + P_{p_k,zP}^{II} - (k - 1)P \quad \text{by (18)} \\ &= n_1 P + n_2 P + \cdots + n_k P - (k - 1)P \quad \text{by (26)} \\ &= (n_1 + n_2 + \cdots + n_k - (k - 1))P. \end{aligned}$$

□

Lemma 23 ([12], Lemma 27).

For $n/m \in \mathbb{Q}_+$ ($n, m \in \mathbb{N}$), let $Q = P_{n/m,zP}^{II}$. Let $n = p_1 p_2 \cdots p_k$ and $m = q_1 q_2 \cdots q_{k'}$ be prime factorizations of n and m , respectively. Let n_i be the second component of the output of Algorithm 3 with inputs (P, z, p_i) , and let m_j be the second component of the output of Algorithm 3 with inputs (P, z, q_j) . Then,

$$Q = ((n_1 + n_2 + \cdots + n_k) - (m_1 + m_2 + \cdots + m_{k'}) - (k - k' - 1))P.$$

Proof. By Lemma 22, we have

$$\begin{aligned} P_{n,zP}^{II} &= (n_1 + n_2 + \cdots + n_k - (k - 1))P, \\ P_{m,zP}^{II} &= (m_1 + m_2 + \cdots + m_{k'} - (k' - 1))P. \end{aligned}$$

Therefore,

$$\begin{aligned} Q &= P_{n/m,zP}^{II} \\ &= P_{n,zP}^{II} - P_{m,zP}^{II} + P \\ &= ((n_1 + n_2 + \cdots + n_k) - (m_1 + m_2 + \cdots + m_{k'}) - (k - k' - 1))P. \end{aligned}$$

□

Theorem 24 ([12], Theorem 28).

For given $P, Q \in E(\mathbb{F}_p)$ and $z \in \mathbb{Z}/l\mathbb{Z}$, suppose that $E(\mathbb{F}_p)$ and $Z = zP$ satisfy one of Conditions 15(a'), (b'), and (c).

(a) For a solution $n/m \in \mathbb{Q}_+$ to $\text{MeDLP}_{II}(P, zP, Q)$, if n and m can be factored, then a solution to $\text{ECDLP}(P, Q)$ can be computed in polynomial time from their prime factorizations.

(b) If there exists an algorithm $\mathcal{A}$ that solves $\text{MeDLP}_{II}(P, zP, Q)$ in polynomial time and the numerator and denominator of the solution can be factored in polynomial time, then $\text{ECDLP}(P, Q)$ can be solved in polynomial time.

Proof. (a) Let $n = p_1 p_2 \cdots p_k$ and $m = q_1 q_2 \cdots q_{k'}$ be prime factorizations of n and m , respectively. Let n_i be the second component of the output of Algorithm 3 with input (P, z, p_i) , and let m_j be the second component of the output of Algorithm 3 with input (P, z, q_j) . Then, by Lemma 23,

$$Q = ((n_1 + n_2 + \cdots + n_k) - (m_1 + m_2 + \cdots + m_{k'}) - (k - k' - 1))P.$$

Therefore, $(n_1 + n_2 + \cdots + n_k) - (m_1 + m_2 + \cdots + m_{k'}) - (k - k' - 1)$ is a solution to $\text{ECDLP}(P, Q)$.

(b) This is a restatement of (a). □

4.5.4 Conclusion of Section 4.5

By Theorem 21, if $E(\mathbb{F}_p)$ and $Z = zP$ satisfy one of Conditions 15(a'), (b'), and (c), then

$$\text{hardness of } \text{MeDLP}_I(P, zP, Q) \geq \text{hardness of } \text{ECDLP}(P, Q). \quad (27)$$

By Theorem 24, if $E(\mathbb{F}_p)$ and $Z = zP$ satisfy one of Conditions 15(a'), (b'), and (c), and if the numerator and denominator of a solution to $\text{MeDLP}_{II}(P, zP, Q)$ can be factored, then

$$\text{hardness of } \text{MeDLP}_{II}(P, zP, Q) \geq \text{hardness of } \text{ECDLP}(P, Q). \quad (28)$$

Remark 25.

For $X \in \{I, II\}$, in order to construct a solution to $\text{ECDLP}(P, Q)$ from a solution to $\text{MeDLP}_X(P, zP, Q)$, z satisfying $Z = zP$ is required. If z is unknown, neither (27) nor (28) can be claimed.

4.6 Consideration of Constructing Solutions to MeDLP from Solutions to ECDLP

The following proposition describes the construction of a solution to MeDLP from a solution to ECDLP.

Proposition 26 ([9], Proposition 12, [12], Proposition 29).

Let $X \in \{I, II\}$. Let $P \in E(\mathbb{F}_p)$ ($P \neq \mathcal{O}$), $l = \#E(\mathbb{F}_p)$, where l is odd, and let $z \in \mathbb{Z}/l\mathbb{Z}$ ($z \neq 0$), and $Z = zP$. Suppose that $\text{ECDLP}(Z, Q - P)$ has a solution m ; that is, $Q - P = mZ$. Define n as follows:

$$n = \begin{cases} m/2 \bmod l & \text{if } \text{sign}(Z) = 1 \\ -m \bmod l & \text{if } \text{sign}(Z) = -1 \end{cases}$$

Then 2^n is a solution to $\text{MeDLP}_X(P, zP, Q)$.

Proof. For $X = I$, by Lemma 16, $Q - P = mZ = \mathcal{O}_{2^n, Z}^I$. Hence, by Theorem 4(a),

$$Q = \mathcal{O}_{2^n, Z}^I + P = P_{2^n, Z}^I.$$

Therefore, 2^n is a solution to $\text{MeDLP}_I(P, zP, Q)$.

For $X = II$, by Lemma 16, $Q - P = mZ = \mathcal{O}_{2^n, Z}^{II}$. Hence, by Theorem 12(a),

$$Q = \mathcal{O}_{2^n, Z}^{II} + P = P_{2^n, Z}^{II}.$$

Therefore, 2^n is a solution to $\text{MeDLP}_{II}(P, zP, Q)$. □

Corollary 27 ([14], Corollary 18).

Let $X \in \{I, II\}$. Let $P \in E(\mathbb{F}_p)$ ($P \neq \mathcal{O}$), $l = \#E(\mathbb{F}_p)$, where l is odd, and let $z \in \mathbb{Z}/l\mathbb{Z}$ ($z \neq 0$), and $Z = zP$. Assume that $\gcd(z, l) = 1$. Suppose that $\text{ECDLP}(P, Q)$ has a solution t . Let

$$m = (t - 1)/z \bmod l \tag{29}$$

and define n as follows:

$$n = \begin{cases} m/2 \bmod l & \text{if } \text{sign}(Z) = 1 \\ -m \bmod l & \text{if } \text{sign}(Z) = -1 \end{cases}$$

Then 2^n is a solution to $\text{MeDLP}_X(P, zP, Q)$.

Proof. By assumption, $Q = tP$, and by (29), $mzP = (t - 1)P$ and hence $mZ = tP - P = Q - P$. Therefore, Proposition 26 can be applied. □

Remark 28.

Suppose that there exists an attacker capable of solving the ECDLP . Can this attacker solve $\text{MeDLP}_X(P, zP, Q)$ and use the solution to attack a cryptographic protocol whose security is based on the hardness of the MeDLP ? Suppose that this cryptographic protocol uses $\mathbb{F}_p$ with a bit length of several hundred bits. The attacker can obtain a solution t to $\text{ECDLP}(P, Q)$, and by computing n as in Corollary 27, 2^n is a solution to $\text{MeDLP}_X(P, zP, Q)$. However, except in very special cases, n has a bit length of several hundred bits; that is, the bit length of 2^n is exponential in the bit length of n . Therefore, directly using the solution 2^n to attack the cryptographic protocol may require exponential time.

4.7 Nonperiodicity of the 2nd Me-Scalar Multiplication

Proposition 29 ([14], Proposition 26).

Let $P, Z \in E(\mathbb{F}_p)$ ($Z \neq \mathcal{O}$). Consider the sequence $\{P_{n, Z}^{II} : n = 1, 2, 3, \dots\}$. Suppose that there exist $k, l \in \mathbb{N}$ such that $P_{i+k} = P_i$ for all $i \geq l$. (That is, suppose that this sequence is eventually periodic.)

$$\begin{array}{ccccccc} P_{1, Z}^{II} & \rightarrow & P_{2, Z}^{II} & \rightarrow & \dots & \rightarrow & P_{l-1, Z}^{II} & \rightarrow & P_{l, Z}^{II} & \rightarrow & P_{l+1, Z}^{II} \\ & & & & & & \parallel & & \searrow & & \\ & & & & & & P_{l+k, Z}^{II} & & & & P_{l+2, Z}^{II} \\ & & & & & & \nwarrow & & \downarrow & & \\ & & & & & & P_{l+k-1, Z}^{II} & \leftarrow & \dots & & \end{array}$$

Then, $P_{n, Z}^{II} = P$ holds for all $n \in \mathbb{Q}_+$.

Proof. Let $t_0 \in \mathbb{N}$ be the smallest natural number satisfying $t_0 \geq l/k$. Since $t_0 k \geq l$, the assumption with $i = t_0 k$ gives

$$P_{(t_0+1)k,Z}^{II} = P_{t_0 k,Z}^{II}.$$

Furthermore, since $(t_0 + 1)k \geq l$,

$$P_{(t_0+2)k,Z}^{II} = P_{(t_0+1)k,Z}^{II}.$$

Repeating this argument, we obtain

$$P_{tk,Z}^{II} = P_{t_0 k,Z}^{II} \text{ for all } t \geq t_0. \quad (30)$$

For any $n = n_0/n_1 \in \mathbb{Q}_+$ ($n_0, n_1 \in \mathbb{N}$), we can choose $m \in \mathbb{N}$ such that $mn_0, mn_1 \geq t_0$. By (30),

$$P_{mn_0 k,Z}^{II} = P_{mn_1 k,Z}^{II} = P_{t_0 k,Z}^{II}. \quad (31)$$

Therefore,

$$\begin{aligned} P_{n,Z}^{II} &= P_{n_0/n_1,Z}^{II} \\ &= P_{(mn_0 k)/(mn_1 k),Z}^{II} && \text{by Proposition 10} \\ &= P_{mn_0 k,Z}^{II} - P_{mn_1 k,Z}^{II} + P && \text{by Definition 9} \\ &= P. && \text{by (31)} \end{aligned}$$

Thus, the proposition is proved. $\square$

Corollary 30 ([14], Corollary 27).

If there exist $i, j \in \mathbb{N}$ such that $P_{i,Z}^{II} \neq P_{j,Z}^{II}$, the sequence $\{P_{n,Z}^{II} : n = 1, 2, 3, \dots\}$ is not eventually periodic.

Proof. This follows from the contrapositive of Proposition 29. $\square$

Corollary 31 ([14], Corollary 28).

For given $P \in E(\mathbb{F}_p)$ and $z \in \mathbb{Z}$, suppose that $E(\mathbb{F}_p)$ and Z satisfy one of Conditions 15(a), (b), and (c). Then, the sequence $\{P_{n,Z}^{II} : n = 1, 2, 3, \dots\}$ is not eventually periodic.

Proof. By Proposition 17, there exist $i, j \in \mathbb{N}$ such that $P_{i,Z}^{II} \neq P_{j,Z}^{II}$. Therefore, the corollary follows from Corollary 30. $\square$

5 Conclusion and Future Work

In this paper, we defined the Me operation, Me-scalar multiplication, and the MeDLP, and proved several properties of the Me operation and Me-scalar multiplication. Furthermore, we showed that, under certain conditions, the MeDLP is as difficult as or more difficult than the ECDLP. We also showed that the sequence $\{P_{n,Z}^{II} : n = 1, 2, 3, \dots\}$ has no period unless it is constant.

The Me operation and Me-scalar multiplication have properties that appear useful for constructing cryptographic protocols. However, since it remains unclear whether the MeDLP is sufficiently difficult and the MeCDH problem can be solved easily, it is difficult to apply the Me operation to many DH-based cryptographic protocols. The construction of cryptographic protocols using the Me operation is left for future work.

Since Shor's algorithm is a quantum algorithm that exploits periodicity over groups, the nonperiodicity of the sequence $\{P_{n,Z}^{II} : n = 1, 2, 3, \dots\}$ is an interesting property when considering quantum algorithms for the MeDLP. Investigating quantum algorithms for the MeDLP, particularly whether a Shor-type algorithm can be applied, is an important direction for future work.

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A M-Scalar Multiplication and MDLP

In this appendix, we define M-scalar multiplication, the MDLP, and the MCDH problem, and then show that, unlike the MeDLP, the MDLP cannot be considered difficult³. The essential argument is the same as that in Sections 4.2.1, 4.3, 4.5.1, and 4.5.2. Although the existence of solutions to the MDLP is not considered here, an argument similar to that in Section 4.4 may be possible.

M-scalar multiplication corresponds to the 1st Me-scalar multiplication. The reason why the author did not consider 2nd M-scalar multiplication is that the MDLP cannot be considered difficult and thus cannot be applied to cryptography. Note that M-scalar multiplication, like Me-scalar multiplication, is not difficult at all.

A.1 M-Scalar Multiplication

By Proposition 1(a), for $a \in \mathbb{F}_p$,

$$(\cdots (((a \oplus a) \oplus a) \oplus a) \cdots \oplus a) = a.$$

Hence, simple repeated applications of $\oplus$ are meaningless. We thus define the M-scalar multiplication $a_{n,z}$ of a with auxiliary element $z \in \mathbb{F}_p$ as the output of Algorithm 4. Note that when $n = 1$, Steps 2 through 5 are not executed, and $a_{1,z} = a$.

Algorithm 4 (Computation of the M-Scalar Multiplication)

Input: $a, z \in \mathbb{F}_p, n = (1, n_{t-2}, \dots, n_0)_2 \in \mathbb{N}$

Output: $a_{n,z} \in \mathbb{F}_p$

1. $x \leftarrow a$
 2. **for** $i = t - 2$ **down to** 0
 3. $x \leftarrow (z + x) \oplus x$
 4. **if** $n_i = 1$ **then** $x \leftarrow x \oplus a$
 5. **end for**
 6. **return** x
-

The M operation and M-scalar multiplication have the following properties.

Theorem 32 ([4]).

Let $a, b, z \in \mathbb{F}_p$ and $n, m \in \mathbb{N}$. Then the following hold.

- (a) $a_{n,z} + b = a + b_{n,z} = (a + b)_{n,z}$.
- (b) $(a_{n,z})_{m,z} = (a_{m,z})_{n,z} = a_{n,z} + a_{m,z} - a$.
- (c) $a_{n,z} \oplus b_{n,z} = (a \oplus b)_{n,z}$.

The following two lemmas are needed in the proof of Theorem 32.

Lemma 33 ([4]).

For $a, b, z \in \mathbb{F}_p$, $(z + (a \oplus b)) \oplus (a \oplus b) = ((z + a) \oplus a) \oplus ((z + b) \oplus b)$ holds.

Proof. We obtain the following calculations, which show that the left-hand side is equal to the

³No efficient algorithm for solving the MDLP has been found.

right-hand side.

$$\begin{aligned}
\text{LHS} &= (z + (a \oplus b)) \oplus (a \oplus b) \\
&= ((z + a) \oplus (z + b)) \oplus (a \oplus b) && \text{by Proposition 1(c)} \\
&= M(M(z + a, z + b), M(a, b)) \\
&= M(M(a - b, 0) + z + b, M(a - b, 0) + b) && \text{by Proposition 1(c)} \\
&= M((M(a - b, 0) + b) + z, M(a - b, 0) + b) \\
&= M(M(a - b, 0) + z, M(a - b, 0)) + b && \text{by Proposition 1(c)} \\
&= M(z, 0) + M(a - b, 0) + b, && \text{by Proposition 1(c)}
\end{aligned}$$

and

$$\begin{aligned}
\text{RHS} &= ((z + a) \oplus a) \oplus ((z + b) \oplus b) \\
&= M(M(z + a, a), M(z + b, b)) \\
&= M(M(z, 0) + a, M(z, 0) + b) && \text{by Proposition 1(c)} \\
&= M((M(z, 0) + a - b) + b, M(z, 0) + b) \\
&= M(M(z, 0) + a - b, M(z, 0)) + b && \text{by Proposition 1(c)} \\
&= M(a - b, 0) + M(z, 0) + b. && \text{by Proposition 1(c)}
\end{aligned}$$

□

Lemma 34.

For $a, z \in \mathbb{F}_p$ and $n \in \mathbb{N}$,

$$a_{2n,z} = (z + a_{n,z}) \oplus a_{n,z}.$$

Proof. Let the binary representation of n be $(1, n_{l-2}, n_{l-3}, \dots, n_0)_2$. Then the binary representation of $2n$ is $(1, n_{l-2}, n_{l-3}, \dots, n_0, 0)_2$. When $a_{2n,z}$ is computed using Algorithm 4, the value of x at the beginning of the last iteration is $a_{n,z}$. Therefore, in Step 3, $x \leftarrow (z + a_{n,z}) \oplus a_{n,z}$ is computed. Since the condition of the if statement in Step 4 is false, Step 4 is not executed, and $(z + a_{n,z}) \oplus a_{n,z}$ is returned in Step 6. Hence, $a_{2n,z} = (z + a_{n,z}) \oplus a_{n,z}$. □

Proof of Theorem 32.

(a) In the computation of $a_{n,z}$ using Algorithm 4, let $x[a_{n,z}]_{j,s}$ denote the value of x after Step s is executed for $i = j$. We consider the following equations:

$$\begin{aligned}
x[a_{n,z}]_{t-2,3} + b &= x[b_{n,z}]_{t-2,3} + a \\
&= x[(a + b)_{n,z}]_{t-2,3},
\end{aligned} \tag{32}$$

$$\begin{aligned}
x[a_{n,z}]_{j,3} + b &= x[b_{n,z}]_{j,3} + a \\
&= x[(a + b)_{n,z}]_{j,3},
\end{aligned} \tag{33}$$

$$\begin{aligned}
x[a_{n,z}]_{j,4} + b &= x[b_{n,z}]_{j,4} + a \\
&= x[(a + b)_{n,z}]_{j,4},
\end{aligned} \tag{34}$$

and

$$\begin{aligned}
x[a_{n,z}]_{j-1,3} + b &= x[b_{n,z}]_{j-1,3} + a \\
&= x[(a + b)_{n,z}]_{j-1,3}.
\end{aligned} \tag{35}$$

For (32)–(35), to prove (a), it suffices to show the following:

- (i) Equation (32) holds.
- (ii) If (33) holds, then (34) holds.

(iii) If (34) holds, then (35) holds.

(i) By direct computation, we have

$$\begin{aligned}
x[a_{n,z}]_{t-2,3} + b &= ((z + a) \oplus a) + b \\
&= ((z + a) + b) \oplus (a + b) \quad \text{by Proposition 1(c)} \\
&= (z + (a + b)) \oplus (a + b), \\
x[b_{n,z}]_{t-2,3} + a &= ((z + b) \oplus b) + a \\
&= ((z + b) + a) \oplus (b + a) \quad \text{by Proposition 1(c)} \\
&= (z + (a + b)) \oplus (a + b),
\end{aligned}$$

and

$$x[(a + b)_{n,z}]_{t-2,3} = (z + (a + b)) \oplus (a + b).$$

Thus, (32) holds.

(ii) If $n_i = 0$, Step 4 of Algorithm 4 does nothing, and hence (33) and (34) are identical. Thus, (ii) holds.

If $n_i = 1$, let A denote the value in (33). Then, we have

$$\begin{aligned}
x[a_{n,z}]_{j,3} &= A - b \\
x[b_{n,z}]_{j,3} &= A - a \\
x[(a + b)_{n,z}]_{j,3} &= A.
\end{aligned}$$

Therefore, we obtain

$$\begin{aligned}
x[a_{n,z}]_{j,4} + b &= ((A - b) \oplus a) + b \\
&= ((A - b) + b) \oplus (a + b) \quad \text{by Proposition 1(c)} \\
&= A \oplus (a + b), \\
x[b_{n,z}]_{j,4} + a &= ((A - a) \oplus b) + a \\
&= ((A - a) + a) \oplus (b + a) \quad \text{by Proposition 1(c)} \\
&= A \oplus (a + b),
\end{aligned}$$

and

$$x[(a + b)_{n,z}]_{j,4} = A \oplus (a + b).$$

Thus, (ii) holds.

(iii) Let B denote the value in (34). Then we have

$$\begin{aligned}
x[a_{n,z}]_{j,4} &= B - b \\
x[b_{n,z}]_{j,4} &= B - a \\
x[(a + b)_{n,z}]_{j,4} &= B.
\end{aligned}$$

Therefore, we have

$$\begin{aligned}
x[a_{n,z}]_{j-1,3} + b &= (z + B - b) \oplus (B - b) + b \\
&= (z + B - b + b) \oplus (B - b + b) \quad \text{by Proposition 1(c)} \\
&= (z + B) \oplus B, \\
x[b_{n,z}]_{j-1,3} + a &= (z + B - a) \oplus (B - a) + a \\
&= (z + B - a + a) \oplus (B - a + a) \quad \text{by Proposition 1(c)} \\
&= (z + B) \oplus B,
\end{aligned}$$

and

$$x[(a+b)_{n,z}]_{j-1,3} = (z+B) \oplus B$$

Hence, (iii) holds.

(b) Using (a), we have

$$\begin{aligned} (a_{n,z})_{m,z} + a &= (a_{n,z} + a)_{m,z} \\ &= (a + a_{n,z})_{m,z} \\ &= a_{m,z} + a_{n,z} \\ &= (a_{m,z})_{n,z} + a. \end{aligned}$$

Hence, (b) holds.

(c) The case $n = 1$ is obvious. We prove (c) for $n \geq 2$ by induction.

When $n = 2$, by the definition of the M-scalar multiplication, we have

$$a_{2,z} \oplus b_{2,z} = ((z+a) \oplus a) \oplus ((z+b) \oplus b), \quad (36)$$

and

$$(a \oplus b)_{2,z} = (z + (a \oplus b)) \oplus (a \oplus b). \quad (37)$$

By Lemma 33, (36) and (37) are equal.

Assume that

$$a_{n,z} \oplus b_{n,z} = (a \oplus b)_{n,z} \quad (38)$$

holds for $n = 2, \dots, k-1$.

If $k \geq 4$ is even, (38) gives

$$a_{k/2,z} \oplus b_{k/2,z} = (a \oplus b)_{k/2,z}. \quad (39)$$

Then, by Lemma 34,

$$a_{k,z} \oplus b_{k,z} = ((z + a_{k/2,z}) \oplus a_{k/2,z}) \oplus ((z + b_{k/2,z}) \oplus b_{k/2,z}), \quad (40)$$

and by (39), we have

$$\begin{aligned} (a \oplus b)_{k,z} &= (z + (a \oplus b)_{k/2,z}) \oplus (a \oplus b)_{k/2,z} \\ &= (z + (a_{k/2,z} \oplus b_{k/2,z})) \oplus (a_{k/2,z} \oplus b_{k/2,z}). \end{aligned} \quad (41)$$

By Lemma 33, (40) and (41) are equal.

If k is odd, (38) gives

$$a_{k-1,z} \oplus b_{k-1,z} = (a \oplus b)_{k-1,z}. \quad (42)$$

By (a),

$$a_{k-1,z} - a = b_{k-1,z} - b.$$

Let this value be x . Then, we have

$$\begin{cases} a_{k-1,z} = x + a, \\ b_{k-1,z} = x + b. \end{cases}$$

Therefore, by Algorithm 4 and Lemma 33, we have

$$\begin{aligned} a_{k,z} \oplus b_{k,z} &= (a_{k-1,z} \oplus a) \oplus (b_{k-1,z} \oplus b) \\ &= ((x+a) \oplus a) \oplus ((x+b) \oplus b) \\ &= (x + (a \oplus b)) \oplus (a \oplus b). \end{aligned} \quad (43)$$

By (a)

$$\underbrace{a_{k-1,z} - a}_{=x} + (a \oplus b) = (a \oplus b)_{k-1,z}.$$

Therefore, by (43),

$$\begin{aligned} a_{k,z} \oplus b_{k,z} &= (x + (a \oplus b)) \oplus (a \oplus b) \\ &= (a \oplus b)_{k-1,z} \oplus (a \oplus b) \\ &= (a \oplus b)_{k,z}. \end{aligned}$$

Thus, (c) holds. $\square$

A.2 MDLP and MCDH Problem

We define M-scalar multiplication versions of the discrete logarithm problem and the CDH problem.

Definition 35.

(a) Given $a, z, b \in \mathbb{F}_p$, the problem of finding an $n \in \mathbb{N}$ satisfying $b = a_{n,z}$ is called the discrete logarithm problem for the M-scalar multiplication (MDLP). We denote this problem by $\text{MDLP}(a, z, b)$.

(b) For $a, z \in \mathbb{F}_p$ and $n, m \in \mathbb{N}$, let $b_1 = a_{n,z}$ and $b_2 = a_{m,z}$. The problem of computing $(a_{n,z})_{m,z}$ from b_1 and b_2 is called the computational Diffie–Hellman problem for the M-scalar multiplication (MCDH problem). We denote this problem by $\text{MCDH}(a, z, b_1, b_2)$.

A.2.1 Hardness of the MDLP

For $\mathcal{X} \in \mathbb{F}_p \times \mathbb{F}_p$, let $\mathcal{X}[1]$ and $\mathcal{X}[2]$ denote the first and second components of $\mathcal{X}$, respectively. Define the operation $+$ on $\mathbb{F}_p \times \mathbb{F}_p$ as

$$\mathcal{X} + \mathcal{Y} = (\mathcal{X}[1] + \mathcal{Y}[1], \mathcal{X}[2] + \mathcal{Y}[2]).$$

For $\mathcal{X}, \mathcal{Y} \in \mathbb{F}_p \times \mathbb{F}_p$, fix $k \in \mathbb{N}$, and define $\mathcal{X} \oplus \mathcal{Y}$ as follows:

$$\mathcal{X} \oplus \mathcal{Y} = \begin{cases} (\mathcal{X}[1], \mathcal{X}[2]) & \text{if } \mathcal{X}[1] = \mathcal{Y}[1] \\ (k\mathcal{X}[1] - (k-1)\mathcal{Y}[1], k\mathcal{X}[2] - (k-1)\mathcal{Y}[2]) & \text{if } \left(\frac{\mathcal{X}[1] - \mathcal{Y}[1]}{p}\right) = 1 \\ (k\mathcal{Y}[1] - (k-1)\mathcal{X}[1], k\mathcal{Y}[2] - (k-1)\mathcal{X}[2]) & \text{if } \left(\frac{\mathcal{X}[1] - \mathcal{Y}[1]}{p}\right) = -1 \end{cases}$$

Considering only the first component, the operation $\oplus$ on $\mathbb{F}_p \times \mathbb{F}_p$ is the same as the operation $\oplus$ on $\mathbb{F}_p$. Therefore,

$$(\mathcal{X} \oplus \mathcal{Y})[1] = \mathcal{X}[1] \oplus \mathcal{Y}[1]. \quad (44)$$

Algorithm 5 takes an element of $\mathbb{F}_p \times \mathbb{F}_p$ as input and has a structure similar to that of Algorithm 4.

Algorithm 5 has the following properties.

Proposition 36 ([8]).

Let $\mathcal{X}'$ be the output of Algorithm 5 with inputs $a(\neq 0), z \in \mathbb{F}_p$, and $n \in \mathbb{N}$. Then, the following hold.

- (a) $\mathcal{X}'[1] = a_{n,z}$.
- (b) $\mathcal{X}'[2] \cdot a = a_{n,z}$.

Algorithm 5

Input: $a(\neq 0), z \in \mathbb{F}_p, n = (1, n_{t-2}, \dots, n_0)_2 \in \mathbb{N}$ **Output:** $\mathcal{X} \in \mathbb{F}_p \times \mathbb{F}_p$

1. $\mathcal{A} = (a, 1)$
 2. $\mathcal{Z} = (z, z/a)$
 3. $\mathcal{X} = \mathcal{A}$
 4. **for** $i = t - 2$ **down to** 0
 5. $\mathcal{X} = (\mathcal{Z} + \mathcal{X}) \oplus \mathcal{X}$
 6. **if** $n_i = 1$ **then** $\mathcal{X} = \mathcal{X} \oplus \mathcal{A}$
 7. **end for**
 8. **return** $\mathcal{X}$
-

Proof. (a) By (44), the processing of the first component in Algorithm 5 with inputs $a, z \in \mathbb{F}_p$ and $n \in \mathbb{N}$ is identical to the processing in Algorithm 4 with inputs $a, z \in \mathbb{F}_p$ and $n \in \mathbb{N}$. Therefore, (a) holds.

(b) The initial values $\mathcal{A}$ and $\mathcal{Z}$ satisfy $\mathcal{A}[1] = \mathcal{A}[2] \cdot a$ and $\mathcal{Z}[1] = \mathcal{Z}[2] \cdot a$. Moreover, from the definitions of $+$ and $\oplus$, these relations are preserved at each step of Algorithm 5. Therefore, at every step of Algorithm 5,

$$\mathcal{X}[1] = \mathcal{X}[2] \cdot a.$$

Hence,

$$\mathcal{X}'[2] \cdot a = \mathcal{X}'[1] = a_{n,z}$$

□

Remark 37.

By Lemma 36, if a solution to $\text{MDLP}(a, z, b)$ exists and $\text{MDLP}(a, z, b)$ can be solved, we can find $t \in \mathbb{F}_p$ such that $ta = b$ using Algorithm 5. In other words, we can compute the division $t = b/a$ from a and b . Therefore, we can only state that

$$\text{hardness of } \text{MDLP}(a, z, b) \geq \text{hardness of division } b/a \text{ in } \mathbb{F}_p.$$

From Theorem 32(b), the solution to $\text{MCDH}(a, z, b_1, b_2)$ is $b_1 + b_2 - a$, which is not hard at all.